Toward Visually Robust Quadruped Parkour via Monocular Depth Estimation

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Problem Definition

What is Quadruped Parkour?

A legged robot autonomously traversing obstacles using **vision** and **agile control**.

Replace: montage of quadruped parkour (ANYmal, Go1/Go2 on obstacles)

Why is This Hard?

1. Agile control

12 joints at 50 Hz from partial observations

Replace: simulation screenshot

2. The sim-to-real vision gap

Training images $\neq$ deployment images

Replace: real-world terrain photo

Our Question

Can **monocular depth estimation** replace a depth camera for quadruped parkour?

Replace: Go2 photo with camera locations

- Unitree Go2: RGB cameras, **no depth sensor**
- Goal: train in estimated depth space, deploy from RGB

Background

RL for Legged Locomotion

- **Hwangbo et al. (2019)** — learned motor skills in sim, transferred to ANYmal hardware
- **Rudin et al. (2022)** — massively parallel training (4096 environments), the `legged_gym` framework

Replace: figure from Hwangbo or Rudin — sim training $\rightarrow$ real robot

Hwangbo et al., Science Robotics 2019; Rudin et al., CoRL 2022

The Teacher–Student Paradigm

- **Chen et al. (2020)** "Learning by Cheating" — privileged distillation for autonomous driving
- **Lee et al. (2020)** — applied to locomotion from proprioceptive history
- **Miki et al. (2022)** — perceptive locomotion with exteroceptive depth on ANYmal

Replace: teacher-student concept diagram from one of these papers

Chen et al., CoRL 2020; Lee et al., Science Robotics 2020; Miki et al., Science Robotics 2022

Vision-Based Parkour

- **Agarwal et al. (2023)** — scandots + depth-based student
- **Cheng et al. (2024)** "Extreme Parkour" — CNN–GRU on real hardware
- **Zhuang et al. (2023)** "Robot Parkour Learning" — depth camera on Go1
- **Hoeller et al. (2024)** "ANYmal Parkour" — hierarchical skill selection

All assume a depth camera at deployment.

Replace: 2×2 grid of images from these four papers

Agarwal et al., CoRL 2023; Cheng et al., ICRA 2024; Zhuang et al., CoRL 2023; Hoeller et al., Science Robotics 2024

Bridging the Sim-to-Real Vision Gap

1. Domain randomization	Vary visual properties during training
2. Generative augmentation	Diffusion models for photorealistic images (LucidSim)
3. Canonical representation	Project both domains to shared depth space via DA2

Replace: three-column visual — DR — generative — canonical representation

Tobin et al., IROS 2017; Yu et al., CoRL 2025

Depth Anything V2 (DA2)

- Vision Transformer (ViT)
- Trained on 142M images
- RGB $\rightarrow$ metric depth map
- Multiple model scales: *small*, *base*, *large*

Yang et al., arXiv 2024

Mathematical Framework

Problem Formulation: POMDP

Partially Observable Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma)$

Observations

- Proprioception $o_{\text{prop}} \in \mathbb{R}^{53}$
  angular vel., gravity, joints, contacts
- Exteroception o_{ext}
  scandots $\mathbb{R}^{132}$ or depth image

Actions

- Joint offsets $a_t \in \mathbb{R}^{12}$
- PD controller: $\tau = K_p(a_t - q) - K_d\dot{q}$

Objective

$$J(\pi) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)\right]$$

PPO: How the Teacher Learns

Clipped surrogate objective

$$L^{\mathrm{CLIP}} = \mathbb{E}_t\Big[\min\Big(r_t\,\hat{A}_t,\;\; \mathrm{clip}(r_t,\,1{-}\epsilon,\,1{+}\epsilon)\;\hat{A}_t\Big)\Big]$$

$r_t = \pi_\theta(a_t|o_t)\,/\,\pi_{\theta_{\mathrm{old}}}(a_t|o_t)$

Clip prevents large, destructive updates

Generalized Advantage Estimation (GAE)

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty}(\gamma\lambda)^l\,\delta_{t+l}, \qquad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

λ interpolates: $\lambda{=}0$ (low variance, high bias) $\leftrightarrow$ $\lambda{=}1$ (high variance, low bias)

Schulman et al., arXiv 2017; Schulman et al., ICLR 2016

DAgger: How the Student Learns

Problem with behavioral cloning: errors compound under distribution shift.

DAgger — student acts, teacher labels, iterate:

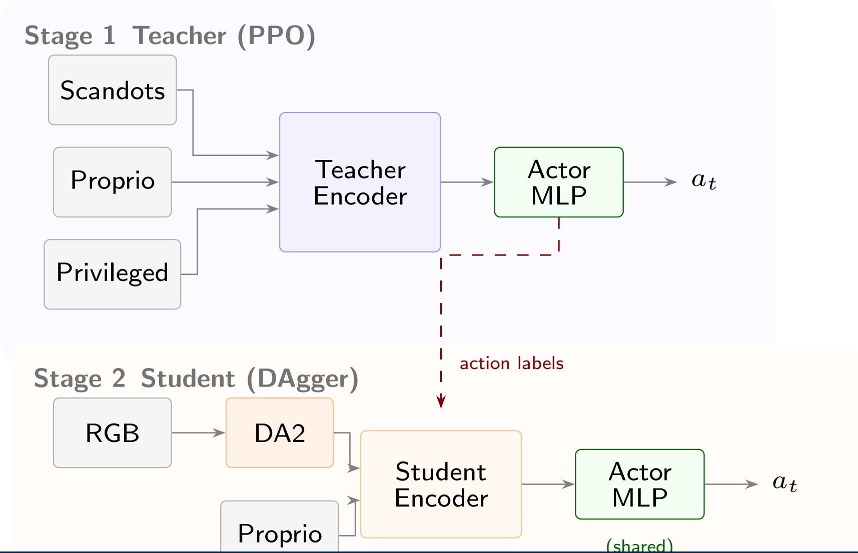
$$\mathcal{L} = \underbrace{\|a_{\text{teacher}} - a_{\text{student}}\|_2}_{\text{action loss}} + \underbrace{\|y_{\text{oracle}} - \hat{y}\|_2}_{\text{heading loss}}$$

The student runs in the environment and collects its own observations,
but the teacher's actions are executed — eliminating distribution mismatch.

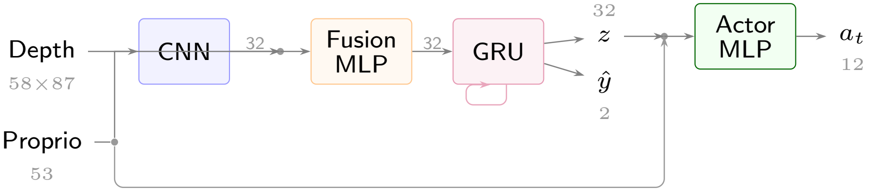
Ross et al., AISTATS 2011

System Architecture

Two-Stage Pipeline

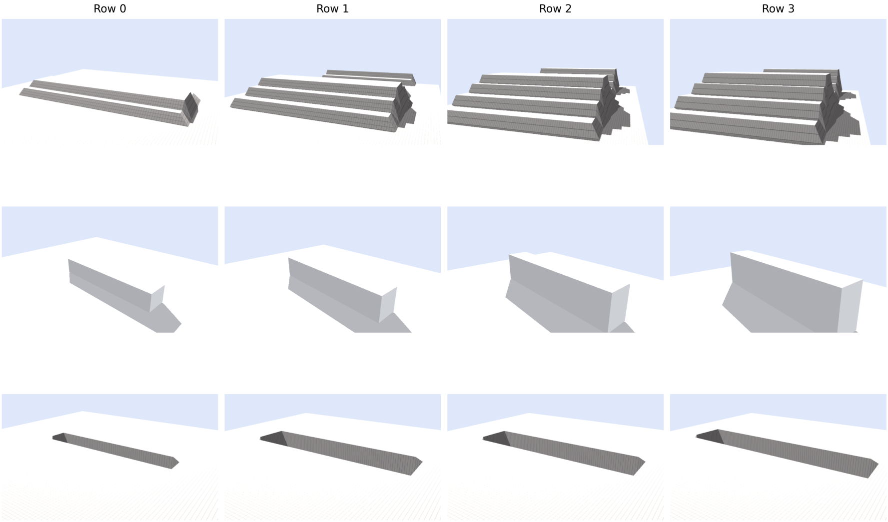


Student Visual Encoder



Depth CNN → Fusion MLP → GRU → terrain latent $z \in \mathbb{R}^{32}$ + heading $\hat{y} \in \mathbb{R}^2$

Terrain and Curriculum



Simulation Demo

Video demo — show simulation runs here

Experiments & Results

Baselines: Upper Bounds

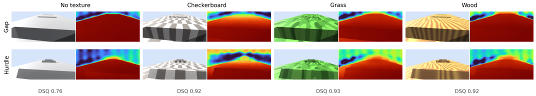
	Row 0	Row 1	Row 2	Row 3
Scandots teacher	98.5	100.0	99.0	98.5
GT-depth student	94.5	94.5	89.1	87.5
Zero-depth ablation	8.0	2.0	0.0	0.0

The student architecture works.
Depth quality is the bottleneck, not the network.

Estimated Depth: Model Scale and Texture

	R0	R1	R2	R3
DA2-small	56	34	—	—
DA2-base	68	83	27	1
DA2-base+tex	65	64	68	45
GT-depth	95	95	89	88

Larger model + texture
→ closes gap to GT-depth



Depth Signal Quality (DSQ)

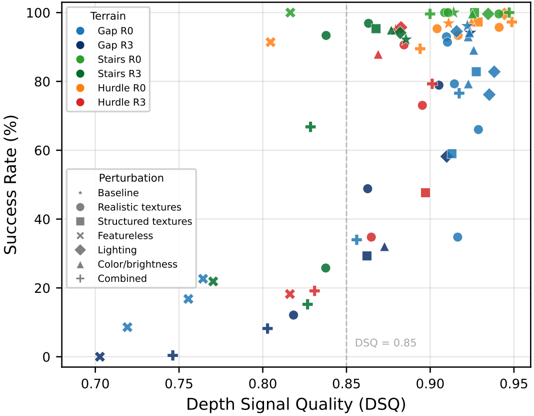
A diagnostic metric: how well does estimated depth preserve terrain geometry?

$$\mathrm{DSQ} = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{N}\sum_{i=1}^{N}\rho\big(d_i^{\mathrm{GT}},\ d_i^{\mathrm{DA2}}\big)$$

Pearson correlation is **affine invariant**: $\rho(\alpha x + \beta,\, y) = \rho(x, y)$

$\rightarrow$ naturally handles monocular scale–shift ambiguity

Visual Sensitivity



- Stairs/hurdles: robust (strong geometric edges)
- Gaps: fragile (boundary depends on texture)
- Featureless $\rightarrow$ catastrophic
- Lighting/color $\rightarrow$ moderate

DSQ predicts failure well,
but high DSQ $\neq$ guaranteed success

Takeaways

What We Learned

1. Monocular depth is a **plausible bridge** from RGB to locomotion
2. Bottleneck = **depth quality**, not student architecture
3. Robust to photometric variation; **fragile on featureless surfaces**

Future work

- Real-world transfer on Unitree Go2
- Temporally consistent depth models
- Joint adaptation of depth estimator + student policy

Thank you!

Questions?